

The Hull–White model under volatility uncertainty

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We study the Hull–White model for the term structure of interest rates in the presence of volatility uncertainty. The uncertainty about the volatility is represented by a set of beliefs, which naturally leads to a sublinear expectation and a G -Brownian motion. The main question in this setting is how to find an arbitrage-free term structure. This question is crucial, since we can show that the classical approach, martingale modeling, does not work in the presence of volatility uncertainty. Therefore, we need to adjust the model in order to find an arbitrage-free term structure. The resulting term structure is affine with respect to the short rate and the adjustment factor. Although the adjustment changes the structure of the model, it is still consistent with the traditional Hull–White model after fitting the yield curve. In addition, we extend the model and the results to a multifactor version, driven by multiple risk factors.

Keywords: Robust finance; Model uncertainty; Interest rates; No-arbitrage

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1. Introduction

Traditional models in finance are subject to model uncertainty and, especially, volatility uncertainty. These models typically assume that there is a single, known probability measure determining the behavior of the fundamental quantities on the market. This assumption simplifies the modeling of financial markets and the pricing of derivatives, but in many situations, it is not possible to specify the underlying probability measure of the model. The uncertainty about using the correct probabilistic law is called *model uncertainty*. There is an increasing literature, commonly referred to as *robust finance*, dealing with model uncertainty by investigating financial markets in the presence of a family of possible probability measures or none at all. The overall aim is to make models robust with respect to misspecifications regarding the probabilistic law. An important kind of model uncertainty is volatility uncertainty. The future evolution of the volatility of an underlying is impossible to predict and it is hard to specify its probabilistic law.

In the present paper, we study the Hull–White model for the term structure of interest rates under volatility uncertainty. That means, we model the instantaneous spot interest rate, called short rate, in the presence of a family of possible beliefs about the volatility and it is completely uncertain which one is correct. We do not impose any probabilistic assumptions on

the beliefs about the volatility, that is, there is no assumption on how likely it is for each belief to be the correct one. We consider all beliefs in which the volatility is bounded by two extreme values. This assumption might be restrictive, but it is sufficient to make the mathematical analysis work. In general, one can relax this assumption with further technical effort and, from an economic perspective, the succeeding results do not require this assumption, as the results do not alter if the bounds for the volatility do.

Mathematically speaking, we model the short rate as an Ornstein–Uhlenbeck process driven by a G (geometric)-Brownian motion. As in the classical Hull–White model, we describe the evolution of the short rate by a diffusion process of Ornstein–Uhlenbeck type. The difference compared to the traditional model is that the volatility is uncertain. We represent the uncertainty about the volatility by a set of probability measures, called the *set of beliefs*, where each measure corresponds to a different belief about the volatility. That means, the short rate has a different volatility under each measure and it is uncertain which measure is the correct belief about the volatility. Such a set of measures naturally leads to a sublinear expectation and a G -Brownian motion. A sublinear expectation can be interpreted as a worst-case measure. A G -Brownian motion is basically a Brownian motion with an uncertain volatility. Thus, we use the G -Brownian motion to describe the evolution of the short rate, i.e. as the driver of the short rate dynamics. Due to the uncertainty about its volatility, the short rate has an uncertain variance.

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The main question in this paper is how to find an arbitrage-free term structure in the presence of volatility uncertainty. The crucial characteristic of volatility uncertainty is that it is represented by a nondominated set of beliefs. That means, there is no measure dominating all measures in the set of beliefs. Hence, it is not possible to find a single equivalent martingale measure for the related bond market. Thus, the discussion about arbitrage becomes a subtle issue in this framework. If we want to follow a martingale modeling approach, we need to choose the bond prices in such a way that the discounted bonds are symmetric G -martingales, which means that they are martingales in each possible scenario for the volatility. Martingale modeling is a common approach in short rate models, but we can unfortunately show that this approach does not work under the initially given set of beliefs.

In order to find an arbitrage-free term structure, we consider sublinear expectations defined by a linear G -backward stochastic differential equation. By standard results on G -backward stochastic differential equations, we can define consistent sublinear expectations by this procedure. Since the G -backward stochastic differential equation is linear, there exists an explicit solution. The representation of the solution shows that the resulting sublinear expectation corresponds to the expectation under an equivalent change of measure. We can also formally show that sublinear expectations defined in this way are in some sense equivalent to the initial one. As a consequence, the bond market is arbitrage-free if there exists a sublinear expectation of this particular type under which the discounted bonds are symmetric G -martingales.

We show that there exists a sublinear expectation of the above kind under which there is a unique arbitrage-free term structure. If we choose a particular process as a coefficient in the linear G -backward stochastic differential equation defining equivalent sublinear expectations, we obtain a sublinear expectation under which there is a unique expression for the bond prices such that the discounted bonds are symmetric G -martingales. The choice might seem special, but it can be justified by economic arguments. Due to the Girsanov transformation for G -Brownian motion, the process represents an adjustment factor, adjusting the short rate by its uncertain variance. Alternatively, we can interpret the process as the market price of risk. Since the model is not only subject to risk but also subject to uncertainty, we also refer to the process as the *market price of uncertainty*. The resulting bond prices are different, though similar, to the prices from the traditional model without volatility uncertainty. In particular, they have an affine structure with respect to the short rate and the market price of uncertainty.

Even though the structure of the model is different from the traditional one, we are yet consistent with the classical Hull–White model after fitting the yield curve. Since we consider an equivalent sublinear expectation, the Girsanov transformation for G -Brownian motion implies that the dynamics of the short rate, as well as the bond prices, differ from the ones of the traditional model. As in the classical model, we use the mean reversion level of the short rate to fit the bond prices of the model to an initial yield curve, observable in the market. Surprisingly, then the short rate dynamics and the bond prices are again consistent with the ones from the classical Hull–White model. They are consistent in the

sense that the short rate dynamics and the bond prices coincide with the classical ones if we drop the uncertainty about the volatility.

In addition, we study an extension of the model driven by multiple risk factors. For the sake of simplicity, we derive the results mentioned above in the presence of a single risk factor, that is, the short rate is driven by a single G -Brownian motion. Such a framework simplifies the interpretation and the intuition of the results and enables us to compare the results with the classical Hull–White model. However, empirical studies show that more factors are needed in order to explain term structure movements (Dai and Singleton 2003, Adrian *et al.* 2013, Joslin *et al.* 2014). Therefore, we consider a model extension in which the short rate is affected by several risk factors with uncertain volatilities and uncertain correlations. We are able to extend all of the previous results to the general case.

Regarding the literature, the concept of model uncertainty originates from *Knightian uncertainty*, which was introduced by Knight (1921). Knightian uncertainty is considered to be the counterpart of risk, because risk can be measured by a probability in contrast to uncertainty. Such uncertainty describes events in reality which are too complex or for which the related data required to calculate the risk of such an event are missing. In such situations, one usually assumes that some quantities are completely uncertain but bounded by two extreme values. In the case of volatility uncertainty, we accordingly consider all volatility scenarios which take values in a certain interval without any further specification.

There are many approaches to describe volatility uncertainty mathematically. On the one hand, there are the approaches of Denis and Martini (2006) and Soner *et al.* (2011), which start from a probabilistic point of view. On the other hand, there is the theory of G -Brownian motion from Peng (2019), in which distributions are characterized by non-linear partial differential equations. Indeed, the approaches of Denis and Martini (2006) and Peng (2019) are connected by a duality, which was shown by Denis *et al.* (2011). Moreover, there are several generalizations, such as an extension to time-dependent, stochastic bounds for the family of volatility scenarios (Nutz 2013) and a pathwise stochastic calculus (Cont and Perkowski 2019, and references therein). We mainly use the theory of G -Brownian motion in this paper, since the literature on G -Brownian motion offers a lot of results.

The literature on model uncertainty in asset markets is very extensive. There are many works on volatility uncertainty in asset markets (Avellaneda *et al.* 1995, Lyons 1995, Epstein and Ji 2013, Vorbrink 2014). Since volatility uncertainty is represented by a nondominated set of probability measures, the classical fundamental theorem of asset pricing cannot be used in such a situation. Therefore, there are many approaches to extend the theorem to a multiprior setting (Bouchard and Nutz 2015, Bayraktar and Zhou 2017, Biagini *et al.* 2017). In addition, there are also model-free versions, i.e. without any reference measure (Riedel 2015, Acciaio *et al.* 2016, Burzoni *et al.* 2019). Another extensively studied topic related to robust finance is the pricing and hedging of contingent claims in the presence of multiple priors (Pos-samaïet *et al.* 2013, Aksamit *et al.* 2019, Carassus *et al.* 2019, and references therein) or without any priors at all (Schied and

Voloshchenko 2016, Beiglböck *et al.* 2017, Bartl *et al.* 2019, and references therein).

The literature on model uncertainty in interest rate models is relatively sparse and there is no paper focusing on obtaining an arbitrage-free term structure in the presence of volatility uncertainty. Epstein and Wilmott (1999) study an interest rate model with no underlying probabilistic assumptions. Instead, they use Knightian uncertainty to describe the evolution of the short rate, which typically leads to a range of prices for contracts on the interest rate. Unfortunately, the discussion about the absence of arbitrage is, like all of the derivations, very intuitive and less mathematical. Avellaneda and Lewicki (1996) adapt the principle of volatility uncertainty to construct an interest rate model that also leads to a range of prices. However, the absence of arbitrage is treated in a very intuitive way as well. Fadina and Schmidt (2019) introduce a term structure model with default risk and ambiguity. The ambiguous parameter is the default intensity and the set of priors is dominated. This simplifies the discussion about arbitrage and results in a range of no-arbitrage prices for defaultable bonds. Fadina *et al.* (2019) study affine processes with parameter uncertainty and the pricing of bonds under ambiguity, which also results in a range of prices. However, this procedure is justified by a superhedging argument from Biagini *et al.* (2017), which is not reasonable for pricing bonds, since they are the fundamental quantities in fixed income markets, i.e. they cannot be hedged.

A different, though related, approach to overcome the stylized facts about the volatility of the short rate is to use regime switching models. Regime switching term structure models assume that the volatility follows a continuous-time Markov chain, which jumps between a finite number of values. The literature on regime switching term structure models shows that these models offer advantages compared to classical term structure models (Monfort and Pegoraro 2007, Dai *et al.* 2007, Gouriéroux *et al.* 2014). In the present paper, we consider all volatility processes bounded by two extreme values. Thus, we also consider trajectories described by regime switching term structure models. However, this is just a pathwise argument. One needs to enlarge the probability space of the model in order to obtain the same Markov chain considered by regime switching term structure models, since the probability of the Markov chain jumping to a different state is usually independent of the remaining risk factors.

In general, volatility uncertainty is a conceptually different approach compared to stochastic volatility. There is plenty of evidence that the volatility of all major financial quantities is neither constant nor deterministic. This problem is addressed by stochastic volatility models, in which the volatility of an underlying is modeled by a particular stochastic process. The latter is typically chosen such that it shares the characteristics of the historical volatility and the model-implied option prices match the prices observed in the market. However, there could be many model specifications performing this task. Moreover, it is not sure if a volatility specification which is consistent with the past is still valid in the future, since the market environment can change drastically. The goal of considering an uncertain volatility is to loosen the assumption that

the probabilistic law of the volatility is perfectly known by considering a family of possible probabilistic laws in order to make the model robust without knowing which one is correct.

The paper is organized as follows. In Section 2, we construct a framework for modeling volatility uncertainty and introduce the short rate process. Section 3 introduces the related bond market. In Section 4, we adapt the concept of martingale modeling to volatility uncertainty and show that martingale modeling does not work in the presence of volatility uncertainty. Hence, we define equivalent sublinear expectations in Section 5, which we can use to find an arbitrage-free term structure. In Section 6, we show that there exists an equivalent sublinear expectation under which we obtain a unique arbitrage-free term structure. Section 7 shows how to fit the model to an initially observed term structure. In Section 8, we extend the model to a version driven by multiple risk factors. Section 9 concludes.

2. Short rate dynamics

In the traditional Hull–White model, without volatility uncertainty, the short rate is described by an Ornstein–Uhlenbeck process driven by a standard Brownian motion. Let us consider a probability space $(\Omega, \mathcal{F}, P_0)$ such that $\Omega = C_0(\mathbb{R}_+)$, $\mathcal{F} = \mathcal{B}(\Omega)$, and P_0 is the Wiener measure. $C_0(\mathbb{R}_+)$ and $\mathcal{B}(\Omega)$ denote the space of all $\mathbb{R}$ -valued continuous paths on $\mathbb{R}_+$ starting in 0 and the Borel σ -algebra on Ω , respectively. We denote by B the canonical process and we choose $(\mathcal{F}_t)_t$ to be the filtration generated by B completed by all P_0 -null sets. Then B is a standard Brownian motion under P_0 . The classical Hull–White model, without volatility uncertainty, assumes that the short rate, denoted by r , satisfies

$$dr_t = (\mu(t) - \alpha r_t) dt + \sigma dB_t, \quad r_0 \in \mathbb{R} \quad (1)$$

for a suitably integrable function $\mu : \mathbb{R}_+ \rightarrow \mathbb{R}$ and constants $\alpha, \sigma > 0$. So the short rate is a mean reverting process with a constant volatility.

In the presence of volatility uncertainty, we consider a family of probability measures, where each measure represents a different belief about the volatility. The state space for the volatility is given by the interval $[\underline{\sigma}, \bar{\sigma}]$ for $\bar{\sigma} \geq \underline{\sigma} > 0$. We denote by $\mathcal{A}$ the collection of all $[\underline{\sigma}, \bar{\sigma}]$ -valued, $(\mathcal{F}_t)_t$ -adapted processes. That is, $\mathcal{A}$ consists of all possible volatility processes bounded by two extreme values. For each $\sigma \in \mathcal{A}$, we define the process B^σ by,

$$B_t^\sigma := \int_0^t \sigma_s dB_s,$$

and we define the measure P^σ to be the law of the process B^σ , i.e.

$$P^\sigma := P_0 \circ (B^\sigma)^{-1}.$$

We call the collection of all such measures $\mathcal{P} := \{P^\sigma \mid \sigma \in \mathcal{A}\}$ the *set of beliefs*, since $\mathcal{P}$ contains all beliefs about the volatility. Now the canonical process has a different volatility under each measure in the set of beliefs. In addition, we define

the sublinear expectation $\hat{\mathbb{E}}$ as the upper expectation of the set of beliefs,

$$\hat{\mathbb{E}}[\xi] := \sup_{P \in \mathcal{P}} \mathbb{E}_P[\xi]$$

for all measurable ξ such that $\mathbb{E}_P[\xi]$ exists for all $P \in \mathcal{P}$. If ξ represents a financial loss, then $\hat{\mathbb{E}}$ can be understood as a risk measure. In general, $\mathbb{E}$ represents a worst-case measure.

Volatility uncertainty naturally leads to a G -Brownian motion. According to Denis *et al.* (2011), $\hat{\mathbb{E}}$ corresponds to the G -expectation on $L_G^1(\Omega)$ and the canonical process B is a G -Brownian motion under $\hat{\mathbb{E}}$. The G -expectation is defined via a nonlinear heat equation, where the nonlinear generator $G : \mathbb{R} \rightarrow \mathbb{R}$ is given by

$$G(a) = \frac{1}{2} \sup_{\sigma \in [\underline{\sigma}^2, \bar{\sigma}^2]} \{\sigma a\}.$$

$L_G^1(\Omega)$ is the space of random variables for which the G -expectation is defined. There are also attempts to enlarge the domain of the G -expectation. Here we use the classical space in order to be able to apply all of the results from the literature on G -Brownian motion. We identify random variables $\xi, \xi' \in L_G^1(\Omega)$ if they are equal *quasi-surely*, i.e. P -almost surely for all $P \in \mathcal{P}$, which is equivalent to $\hat{\mathbb{E}}[|\xi - \xi'|] = 0$. We also use the terminology $\mathcal{P}$ -*quasi-surely* if we need to indicate under which set of measures a statement holds quasi-surely. The details regarding the calculus of G -Brownian motion can be found in the book of Peng (2019).

A G -Brownian motion is tailor-made for modeling volatility uncertainty, since it has no mean uncertainty but variance uncertainty. That means,

$$\begin{aligned} \hat{\mathbb{E}}[B_t] &= 0 = -\hat{\mathbb{E}}[-B_t], \\ \hat{\mathbb{E}}[B_t^2] &= \bar{\sigma}^2 t \geq \underline{\sigma}^2 t = -\hat{\mathbb{E}}[-B_t^2]. \end{aligned}$$

As a consequence, the quadratic variation of a G -Brownian motion is an uncertain process satisfying

$$\bar{\sigma}^2 t \geq \langle B \rangle_t \geq \underline{\sigma}^2 t.$$

Thus, the canonical process is unambiguous in mean and evolves with a volatility being at most $\bar{\sigma}$ and at least $\underline{\sigma}$. The previous inequality shows that B is a standard Brownian motion with a constant volatility if there is no volatility uncertainty, that is, if $\bar{\sigma} = \underline{\sigma}$.

We describe the behavior of the short rate by an Ornstein–Uhlenbeck process driven by a G -Brownian motion. We choose the same structure as in the classical Hull–White model. The difference is that we include volatility uncertainty by replacing the constant volatility and the standard Brownian motion by a G -Brownian motion. Hence, the short rate process r is supposed to be given by the G -stochastic differential equation

$$dr_t = (\mu(t) - \alpha r_t) dt + dB_t, \quad r_0 \in \mathbb{R} \quad (2)$$

for a suitably integrable function $\mu : \mathbb{R}_+ \rightarrow \mathbb{R}$ and a constant $\alpha > 0$. Thus, the short rate has a time-dependent mean reversion level, which is deterministic, and a time-dependent

volatility, which is uncertain. This is desirable, since we can use the mean reversion level for yield curve fitting and we do not have to specify any volatility structure. It should be noted that (2) corresponds to (1), i.e. the classical case without volatility uncertainty, if $\bar{\sigma} = \underline{\sigma} = \sigma$.

The G -stochastic differential equation describing the dynamics of the short rate has a closed-form solution. By Theorem 5.1.3 from Peng (2019), we know that (2) has a unique solution in $M_G^2(0, T)$ for any $T < \infty$. The space $M_G^2(0, T)$ consists of stochastic processes with a certain regularity and is used to define stochastic integrals in the calculus of G -Brownian motion. As in the classical case, we can explicitly solve (2).

PROPOSITION 2.1 *The solution to the G -stochastic differential equation (2) is given by*

$$r_t = e^{-\alpha t} r_0 + \int_0^t e^{-\alpha(t-s)} \mu(s) ds + \int_0^t e^{-\alpha(t-s)} dB_s. \quad (3)$$

Proof This can be verified by using Itô's formula for G -Brownian motion (Li and Peng 2011, Theorem 5.4). The verification works totally analogous to the classical case with a standard Brownian motion. ■

The short rate has no mean uncertainty, but it has an uncertain variance. We can easily show that the upper expectation of the short rate coincides with its lower expectation. Thus, the mean of the short rate is deterministic. In addition, we can show that the upper, respectively lower, expectation of the squared deviation of the short rate from its mean is given by the variance from the classical Hull–White model with the highest, respectively lowest, possible volatility. Hence, the short rate has an uncertain variance, which is bounded by two extreme values.

THEOREM 2.1 *The short rate process r satisfies*

$$\hat{\mathbb{E}}[r_t] = e^{-\alpha t} r_0 + \int_0^t e^{-\alpha(t-s)} \mu(s) ds = -\hat{\mathbb{E}}[-r_t], \quad (4a)$$

$$\begin{aligned} \hat{\mathbb{E}}[(r_t - \hat{\mathbb{E}}[r_t])^2] &= \frac{\bar{\sigma}^2}{2\alpha} (1 - e^{-2\alpha t}) \\ &\geq \frac{\underline{\sigma}^2}{2\alpha} (1 - e^{-2\alpha t}) = -\hat{\mathbb{E}}[-(r_t - \hat{\mathbb{E}}[r_t])^2]. \end{aligned} \quad (4b)$$

Proof First, we sketch how to obtain (4a). The first two summands on the right-hand side of (3) are deterministic. We know that the upper expectation and the lower expectation of an integral with respect to a G -Brownian motion vanish. Thus, it holds (4a).

In order to show (4b), we use the nonlinear Feynman–Kac formula from Hu *et al.* (2014). We define the process X as the deviation of r from its mean, that is,

$$X_t := r_t - \hat{\mathbb{E}}[r_t] = \int_0^t e^{-\alpha(t-s)} dB_s.$$

By Proposition 2.1, we know that X solves the G -stochastic differential equation

$$dX_t = -\alpha X_t dt + dB_t, \quad X_0 = 0.$$

By Theorem 4.4 and Theorem 4.5 of Hu *et al.* (2014), the process Y , defined by $Y_t := \hat{\mathbb{E}}_t[X_T^2]$ for $t \leq T$, is given by $Y_t = u(t, X_t)$, where the function $u : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ is the unique viscosity solution to the nonlinear partial differential equation

$$\partial_t u + G(\partial_{xx}^2 u) - \alpha x \partial_x u = 0, \quad u(T, x) = x^2. \quad (5)$$

One can verify that the solution to (5) is given by

$$u(t, x) = \frac{\sigma^2}{2\alpha} (1 - e^{-2\alpha(T-t)}) + e^{-2\alpha(T-t)} x^2.$$

Thus, we obtain the first equality in (4b). The second follows by the same procedure. ■

3. Related bond market

The corresponding bond market consists of a money-market account and zero-coupon bonds for all possible maturities. First of all, we fix a finite time $\tau < \infty$ and suppose that all trading takes place within the finite time horizon $[0, \tau]$. The market consists of the following investment opportunities. The first one is to invest in the money-market account, which grows by the short rate r . The money-market account is a process denoted by M and is given by

$$M_t := \exp\left(\int_0^t r_s ds\right).$$

In addition to the money-market account, the market offers zero-coupon bonds for all maturities within the time horizon. For $t \leq T \leq \tau$, the price of a bond with maturity T at time t is denoted by $P_t(T)$. The bond has a terminal payoff of 1, that is,

$$P_T(T) = 1.$$

Henceforth, we use the money-market account as a numéraire. That means, we restrict to the discounted bond prices, defined by

$$\tilde{P}_t(T) := M_t^{-1} P_t(T).$$

We assume that, for all $T \leq \tau$, the discounted bond price process $\tilde{P}(T)$ is a diffusion process, i.e.

$$d\tilde{P}_t(T) = \alpha_t(T) dt + \beta_t(T) dB_t + \gamma_t(T) d\langle B \rangle_t$$

for $\alpha(T), \gamma(T), \beta(T) \in M_G^2(0, T)$. This is a technical assumption to make the following definition work. The assumption is satisfied in all of the succeeding scenarios.

The agents can participate in the market by choosing a trading strategy to create a portfolio. Choosing a market strategy means that they can select a finite number of discounted bonds they want to trade and decide on how much of them they want to buy or sell at each time within the time horizon. The value of the related portfolio is the integral of the market strategy with respect to the price processes. That means, we implicitly assume that the trading strategy is self-financing.

DEFINITION 3.1 *An admissible market strategy (π, T) consists of a vector $T \in [0, \tau]^n$ and a bounded process $\pi \in$*

$M_G^2(0, \tau; \mathbb{R}^n)$ for $n \in \mathbb{N}$. The corresponding portfolio value at terminal time is given by

$$\tilde{v}_\tau(\pi, T) := \sum_{i=1}^n \int_0^{T_i} \pi_t^i d\tilde{P}_t(T_i).$$

The restriction to trading a finite number of discounted bonds could be generalized by using methods from large financial markets (Klein *et al.* 2016) or by allowing for continuous trading within the set of all maturities (Björk *et al.* 1997). Here we restrict to trading finitely many discounted bonds, since such a generalization is not the objective of the model.

We use a quasi-sure notion of arbitrage. The classical definition of arbitrage depends on the underlying probability measure of the model. Since we are dealing with more than one measure in the presence of volatility uncertainty, we need to consider a definition slightly different from the classical one. The following definition of arbitrage corresponds to the one that is commonly used in the literature on robust finance (Bouchard and Nutz 2015, for example).

DEFINITION 3.2 *An admissible market strategy (π, T) is called arbitrage strategy if*

$$\tilde{v}_\tau(\pi, T) \geq 0 \text{ quasi-surely,}$$

$$P(\tilde{v}_\tau(\pi, T) > 0) > 0 \text{ for at least one } P \in \mathcal{P}.$$

Moreover, we say that the bond market is arbitrage-free if there is no arbitrage strategy.

This is a weaker version than requiring that the strategy has to be an arbitrage in the classical sense with respect to all measures. The difference is that the probability of a strictly positive win does not have to be strictly positive under each measure.

4. Martingale modeling

Most short rate models use a martingale modeling approach to ensure that the related bond market is arbitrage-free. A standard result in mathematical finance is that the market is arbitrage-free if and only if the traded quantities on the market are martingales under a measure equivalent to the real world measure, termed fundamental theorem of asset pricing. The common practice in short rate models is martingale modeling, since bond markets are incomplete. Incomplete means that there is not a unique martingale measure but many of them. Thus, one usually supposes that the short rate satisfies certain dynamics under a given martingale measure. Then the bond prices are chosen such that the discounted bonds are martingales under the exogenously given martingale measure in order to exclude arbitrage.

In the presence of volatility uncertainty, martingale modeling requires that the discounted bonds are symmetric G -martingales under $\hat{\mathbb{E}}$. If there is volatility uncertainty, the set of beliefs contains mutually singular measures. Hence, there is no dominating measure for the set of beliefs, which implies that it is not possible to find a single martingale measure equivalent to all measures in the set of beliefs. A fundamental theorem of asset pricing under a possibly nondominated

set of beliefs was established by Bouchard and Nutz (2015), for the discrete-time case, and Biagini *et al.* (2017), for the continuous-time case. Roughly speaking, the theorem says that the absence of arbitrage is equivalent to the existence of a set of martingale measures which is in some sense equivalent to the set of beliefs. That means, the price processes have to be martingales under each measure in the equivalent set of measures. Therefore, if we want to follow a martingale modeling approach in the presence of volatility uncertainty, we need to assume that our set of beliefs is a set of exogenously given martingale measures. Then we need to choose the bond prices such that the discounted bonds are martingales under each measure in the set of beliefs. Being a martingale under each measure in the set of beliefs is equivalent to being a symmetric G -martingale under $\hat{\mathbb{E}}$. The sufficiency of this martingale modeling approach for the absence of arbitrage is shown by the following proposition.

PROPOSITION 4.1 *The bond market is arbitrage-free if the discounted bond $\tilde{P}(T)$ is a symmetric G -martingale under $\hat{\mathbb{E}}$ for all $T \leq \tau$.*

Proof Let $\tilde{P}(T)$ be a symmetric G -martingale under $\hat{\mathbb{E}}$ for all $T \leq \tau$ and let us suppose that there is an arbitrage strategy (π, T) . By Definition 3.2, it holds $\tilde{v}_\tau(\pi, T) \geq 0$. Hence, we know that $|\tilde{v}_\tau(\pi, T)| = \tilde{v}_\tau(\pi, T)$, which implies

$$\hat{\mathbb{E}}[|\tilde{v}_\tau(\pi, T)|] = \hat{\mathbb{E}}[\tilde{v}_\tau(\pi, T)]. \quad (6)$$

Using Definition 3.1 and the sublinearity of $\hat{\mathbb{E}}$, we obtain

$$\hat{\mathbb{E}}[\tilde{v}_\tau(\pi, T)] \leq \sum_{i=1}^n \hat{\mathbb{E}} \left[\int_0^{T_i} \pi_t^i d\tilde{P}_t(T_i) \right]. \quad (7)$$

By the representation theorem for symmetric G -martingales (Song 2011, Theorem 4.8), there exists a process $H(T) \in M_G^2(0, T)$ for all $T \leq \tau$ such that

$$d\tilde{P}_t(T) = H_t(T) dB_t.$$

Since π^i is a bounded process in $M_G^2(0, \tau)$, we have $\pi^i H(T_i) \in M_G^2(0, T_i)$ and thus,

$$\hat{\mathbb{E}} \left[\int_0^{T_i} \pi_t^i d\tilde{P}_t(T_i) \right] = \hat{\mathbb{E}} \left[\int_0^{T_i} \pi_t^i H_t(T_i) dB_t \right] = 0 \quad (8)$$

for $i = 1, \dots, n$. Combining (6), (7), and (8), we get $\tilde{v}_\tau(\pi, T) = 0$, which is a contradiction to

$$P(\tilde{v}_\tau(\pi, T) > 0) > 0 \quad \text{for at least one } P \in \mathcal{P}.$$

Therefore, there is no arbitrage strategy. ■

Unfortunately, we can show that the martingale modeling approach in the Hull–White model does not work in the presence of volatility uncertainty. Martingale modeling only works in the classical case when there is no uncertainty about the volatility. In that case the bond prices are obviously given by the bond prices from the classical Hull–White model.

THEOREM 4.1 *Let $T \leq \tau$. Then the discounted bond $\tilde{P}(T)$ is a symmetric G -martingale under $\hat{\mathbb{E}}$ if and only if $\bar{\sigma} = \underline{\sigma}$ and the bond price, for all $t \leq T$, is given by*

$$P_t(T) = \exp(A^\sigma(t, T) - B(t, T)r_t), \quad (9)$$

where the functions $A^\sigma : \mathbb{R}^2 \rightarrow \mathbb{R}$, for $\sigma > 0$, and $B : \mathbb{R}^2 \rightarrow \mathbb{R}$ are defined by

$$A^\sigma(t, T) := \int_t^T \left(\frac{1}{2} \sigma^2 B(s, T)^2 - \mu(s) B(s, T) \right) ds, \quad (10a)$$

$$B(t, T) := \frac{1}{\alpha} (1 - e^{-\alpha(T-t)}), \quad (10b)$$

respectively.

Proof First, let us suppose that the discounted bond $\tilde{P}(T)$ is a symmetric G -martingale under $\hat{\mathbb{E}}$. We show that the expectation of the discount factor is the same under each measure in the set of beliefs. The definition of symmetric G -martingales and the terminal condition of the bond implies

$$\tilde{P}_t(T) = \hat{\mathbb{E}}_t[\tilde{P}(T, T)] = \hat{\mathbb{E}}_t[M_T^{-1}],$$

$$\tilde{P}_t(T) = -\hat{\mathbb{E}}_t[-\tilde{P}(T, T)] = -\hat{\mathbb{E}}_t[-M_T^{-1}].$$

Combining the previous equations and setting $t = 0$ yields

$$\sup_{P \in \mathcal{P}} \mathbb{E}_P[M_T^{-1}] = \inf_{P \in \mathcal{P}} \mathbb{E}_P[M_T^{-1}], \quad (11)$$

which in turn implies that the expectation of M_T^{-1} is the same under each measure.

Now we use the expression for the bond prices from the classical Hull–White model to show that (11) implies $\bar{\sigma} = \underline{\sigma}$ and (9). Let us consider the measures $P^{\bar{\sigma}}, P^{\underline{\sigma}} \in \mathcal{P}$ induced by the highest and the lowest possible volatility, respectively. The expectations of M_T^{-1} under $P^{\bar{\sigma}}$ and $P^{\underline{\sigma}}$ are given by

$$\mathbb{E}_{P^{\bar{\sigma}}}[M_T^{-1}] = \exp(A^{\bar{\sigma}}(0, T) - B(0, T)r_0),$$

$$\mathbb{E}_{P^{\underline{\sigma}}}[M_T^{-1}] = \exp(A^{\underline{\sigma}}(0, T) - B(0, T)r_0),$$

respectively (Björk 2004, Subsection 22.4.4). By (11), the latter expressions are equal. From (10a) and (10b), we see that this only holds if $\bar{\sigma} = \underline{\sigma}$. Since $\bar{\sigma} = \underline{\sigma}$ implies $\mathcal{P} = \{P^{\bar{\sigma}}\}$, we are back in the classical case without volatility uncertainty. In that case, the bond price is given by (9).

Next, let us suppose that $\bar{\sigma} = \underline{\sigma}$ and the bond price is determined by (9). Then we are again back in the classical case without volatility uncertainty and the discounted bond is clearly a martingale under $P^{\bar{\sigma}}$. Since $\mathcal{P} = \{P^{\bar{\sigma}}\}$, the discounted bond is also a symmetric G -martingale. ■

5. Equivalent sublinear expectations

In order to find an arbitrage-free term structure, we consider the following type of sublinear expectations defined by a G -backward stochastic differential equation. Let λ be a bounded

process in $M_G^p(0, \tau)$ for some $p > 1$. We define the sublinear expectation $\tilde{\mathbb{E}}$ on $L_G^p(\Omega_\tau)$ by $\tilde{\mathbb{E}}_t[\xi] := Y_t^\xi$, where Y^ξ solves the G -backward stochastic differential equation

$$-dY_t^\xi = \lambda Z_t dt - Z_t dB_t - dK_t, \quad Y_\tau^\xi = \xi. \quad (12)$$

The reader may refer to the paper of Hu *et al.* (2014) for all details related to G -backward stochastic differential equations.

The representation of the solution to the G -backward stochastic differential equation from above shows that $\tilde{\mathbb{E}}$ is in some sense equivalent to $\hat{\mathbb{E}}$. We can find an explicit solution to (12) if we consider the extended $\tilde{G}$ -expectation space $(\tilde{\Omega}_\tau, L_{\tilde{G}}^1(\tilde{\Omega}_\tau), \tilde{\mathbb{E}})$ with the canonical process $(B, \tilde{B})$, where $\tilde{\Omega}_\tau = C_0^2([0, \tau])$ and the generator $\tilde{G} : \mathbb{S}^2 \rightarrow \mathbb{R}$ is given by

$$\tilde{G}(A) = \frac{1}{2} \sup_{\sigma \in [\underline{\sigma}^2, \bar{\sigma}^2]} \text{tr} \left(\begin{pmatrix} \sigma & 1 \\ 1 & \sigma^{-1} \end{pmatrix} A \right).$$

$C_0^2([0, \tau])$ and $\mathbb{S}^2$ denote the space of all $\mathbb{R}^2$ -valued continuous paths on $[0, \tau]$ starting in 0 and the space of all symmetric 2×2 matrices, respectively. The $\tilde{G}$ -expectation $\tilde{\mathbb{E}}$ is equal to the G -expectation $\hat{\mathbb{E}}$ on the initial domain, that is,

$$\tilde{\mathbb{E}}[\xi] = \hat{\mathbb{E}}[\xi] \quad (13)$$

for all $\xi \in L_G^1(\Omega)$. By Theorem 3.2 of Hu *et al.* (2014), Y^ξ is given by

$$Y_t^\xi = \mathcal{E}_t^{-1} \tilde{\mathbb{E}}_t[\mathcal{E}_\tau \xi],$$

where the process $\mathcal{E}$ is defined by

$$\mathcal{E}_t := \exp \left(\int_0^t \lambda_s d\tilde{B}_s - \frac{1}{2} \int_0^t \lambda_s^2 d\langle \tilde{B} \rangle_s \right),$$

and, by Theorem 5.1 of Hu *et al.* (2014), $\tilde{\mathbb{E}}$ defines a time consistent sublinear expectation. Hence, we see that the definition of $\tilde{\mathbb{E}}$ corresponds to an equivalent change of measure. In Lemma 5.1, we rigorously show that $\tilde{\mathbb{E}}$ is in some sense equivalent to $\hat{\mathbb{E}}$.

There is no arbitrage on the bond market if there exists an equivalent sublinear expectation of the above kind under which the discounted bonds are symmetric G -martingales.

PROPOSITION 5.1 *The bond market is arbitrage-free if the discounted bond $\tilde{P}(T)$ is a symmetric G -martingale under $\tilde{\mathbb{E}}$ for all $T \leq \tau$.*

The proof of Proposition 5.1 relies on the following lemma. The lemma shows that $\tilde{\mathbb{E}}$ is equivalent to $\hat{\mathbb{E}}$ in the sense that the null spaces induced by the natural norms related to $\tilde{\mathbb{E}}$ and $\hat{\mathbb{E}}$ are the same.

LEMMA 5.1 *Let $\xi \in L_G^p(\Omega_\tau)$. Then $\xi = 0$ if and only if $\tilde{\mathbb{E}}[|\xi|] = 0$.*

Proof Let $\xi \in L_G^p(\Omega_\tau)$. From (13), we know that $\xi = 0$ if and only if $\hat{\mathbb{E}}[|\xi|] = 0$. Hence, it suffices to show that $\tilde{\mathbb{E}}[|\xi|] = 0$ if and only if $\hat{\mathbb{E}}[|\xi|] = 0$.

We show the latter by representing $\tilde{\mathbb{E}}$ and $\hat{\mathbb{E}}$ as an upper expectation of a family of probability measures. As in the

beginning of Section 2, we can construct a family $\tilde{\mathcal{P}}$ of probability measures on $(\tilde{\Omega}_\tau, \mathcal{B}(\tilde{\Omega}_\tau))$ such that

$$\tilde{\mathbb{E}}[\xi] = \sup_{\tilde{P} \in \tilde{\mathcal{P}}} \mathbb{E}_{\tilde{P}}[\xi]$$

for all $\xi \in L_{\tilde{G}}^1(\tilde{\Omega}_\tau)$. Moreover, $\mathcal{E}$ solves the G -stochastic differential equation

$$d\mathcal{E}_t = \lambda_t \mathcal{E}_t d\tilde{B}_t, \quad \mathcal{E}_0 = 1.$$

This implies that $\mathcal{E}$ is a symmetric G -martingale, satisfying $\hat{\mathbb{E}}[\mathcal{E}_\tau] = 1$. Thus, for $\tilde{P} \in \tilde{\mathcal{P}}$, we can define a probability measure on $(\tilde{\Omega}_\tau, \mathcal{B}(\tilde{\Omega}_\tau))$ by $Q(\tilde{P}) := \mathcal{E}_\tau \cdot \tilde{P}$. Since $\mathcal{E}_\tau > 0$ $\tilde{P}$ -quasi-surely, we know that $Q(\tilde{P}) \sim \tilde{P}$. If we now define $\mathcal{Q} := \{Q(\tilde{P}) | \tilde{P} \in \tilde{\mathcal{P}}\}$, we obtain

$$\tilde{\mathbb{E}}[\xi] = \sup_{Q \in \mathcal{Q}} \mathbb{E}_Q[\xi]$$

for all $\xi \in L_G^p(\Omega_\tau)$. Since $\mathcal{Q}$ consists of equivalent measures, we get that $\xi = 0$ $\tilde{P}$ -quasi-surely if and only if $\xi = 0$ Q -quasi-surely. Hence, the proof is complete. ■

Using Lemma 5.1 and the Girsanov transformation for G -Brownian motion from Hu *et al.* (2014), we can proceed as in the proof of Proposition 4.1 to prove Proposition 5.1.

Proof of Proposition 5.1 Let $\tilde{P}(T)$ be a symmetric G -martingale under $\tilde{\mathbb{E}}$ for all $T \leq \tau$ and let us suppose that there exists an arbitrage strategy (π, T) . By Definition 3.2, it holds $\tilde{v}_\tau(\pi, T) \geq 0$, which implies $|\tilde{v}_\tau(\pi, T)| = \tilde{v}_\tau(\pi, T)$. By Lemma 5.1, we have

$$\tilde{\mathbb{E}}[|\tilde{v}_\tau(\pi, T)|] = \tilde{\mathbb{E}}[\tilde{v}_\tau(\pi, T)]. \quad (14)$$

Using Definition 3.1 and the sublinearity of $\tilde{\mathbb{E}}$, we obtain

$$\tilde{\mathbb{E}}[\tilde{v}_\tau(\pi, T)] \leq \sum_{i=1}^n \tilde{\mathbb{E}} \left[\int_0^{T_i} \pi_t^i d\tilde{P}_t(T_i) \right]. \quad (15)$$

By Theorem 5.2 of Hu *et al.* (2014), the process $\bar{B}$, defined by

$$\bar{B}_t := B_t - \int_0^t \lambda_s ds,$$

is a G -Brownian motion under $\tilde{\mathbb{E}}$. Since $\tilde{P}(T)$ is a symmetric G -martingale under $\tilde{\mathbb{E}}$ for all $T \leq \tau$, there exists a process $H(T) \in M_G^2(0, T)$ for all $T \leq \tau$ such that

$$d\tilde{P}_t(T) = H_t(T) d\bar{B}_t.$$

Thus, as in the proof of Proposition 4.1, we have

$$\tilde{\mathbb{E}} \left[\int_0^{T_i} \pi_t^i d\tilde{P}_t(T_i) \right] = \tilde{\mathbb{E}} \left[\int_0^{T_i} \pi_t^i H_t(T_i) d\bar{B}_t \right] = 0 \quad (16)$$

for $i = 1, \dots, n$. Combining (14), (15), and (16), we get $\tilde{v}_\tau(\pi, T) = 0$ by Lemma 5.1, which is a contradiction to

$$P(\tilde{v}_\tau(\pi, T) > 0) > 0 \quad \text{for at least one } P \in \mathcal{P}.$$

Therefore, there is no arbitrage strategy. ■

6. Arbitrage-free term structure

There exists an equivalent sublinear expectation of the above kind under which the discounted bonds are symmetric G -martingales. Let us define the process q by

$$q_t := \int_0^t e^{-2\alpha(t-s)} d\langle B \rangle_s.$$

By applying Itô's formula for G -Brownian motion, we observe that q satisfies

$$dq_t = d\langle B \rangle_t - 2\alpha q_t dt, \quad q_0 = 0. \quad (17)$$

If we use the process q to define an equivalent sublinear expectation as in Section 5, we obtain a sublinear expectation under which there is a unique expression for the bond prices such that the discounted bonds are symmetric G -martingales. A justification for choosing the process q is given below.

THEOREM 6.1 *Let $T \leq \tau$ and $\lambda = q$. Then the discounted bond $\tilde{P}(T)$ is a symmetric G -martingale under $\tilde{\mathbb{E}}$ if and only if the bond price, for all $t \leq T$, is given by*

$$P_t(T) = \exp \left(A(t, T) - B(t, T)r_t - \frac{1}{2}B(t, T)^2 q_t \right), \quad (18)$$

where the functions $A : \mathbb{R}^2 \rightarrow \mathbb{R}$ and $B : \mathbb{R}^2 \rightarrow \mathbb{R}$ are defined by

$$A(t, T) := - \int_t^T B(s, T) \mu(s) ds \quad (19)$$

and (10b), respectively.

Proof Let $T \leq \tau$ and $\lambda = q$. First of all, we show that the process X , defined by

$$X_t := \exp \left(A(t, T) - B(t, T)r_t - \frac{1}{2}B(t, T)^2 q_t - \int_0^t r_s ds \right),$$

is a symmetric G -martingale under $\hat{\mathbb{E}}$. Applying Itô's formula for G -Brownian motion to X yields

$$dX_t = \Delta_t X_t dt - B(t, T) X_t dB_t + \Delta'_t X_t d\langle B \rangle_t,$$

where the drift terms Δ and Δ' are given by

$$\begin{aligned} \Delta_t &:= \partial_t A(t, T) - \partial_t B(t, T)r_t - B(t, T)\partial_t B(t, T)q_t \\ &\quad - B(t, T)(\mu(t) - \alpha r_t) + B(t, T)^2 \alpha q_t - r_t \\ &= (\partial_t A(t, T) - \mu(t)B(t, T)) \\ &\quad - (\partial_t B(t, T) - \alpha B(t, T) + 1)r_t \\ &\quad - B(t, T)(\partial_t B(t, T) - \alpha B(t, T))q_t, \\ \Delta'_t &:= -\frac{1}{2}B(t, T)^2 + \frac{1}{2}B(t, T)^2 = 0, \end{aligned}$$

respectively. The functions A and B satisfy

$$\begin{aligned} \partial_t A(t, T) &= \mu(t)B(t, T), \\ \partial_t B(t, T) &= \alpha B(t, T) - 1, \end{aligned}$$

respectively. Thus, we get

$$dX_t = B(t, T)X_t q_t dt - B(t, T)X_t dB_t. \quad (20)$$

Since (20) is a linear G -stochastic differential equation with bounded coefficients, it has a unique solution, which is in $M_G^2(0, T)$. Therefore, $X \in M_G^2(0, T)$, which implies $X_t \in L_G^2(\Omega_t)$ for all $t \leq T$. By the Girsanov transformation for G -Brownian motion, the process $\bar{B}$, defined by

$$\bar{B}_t := B_t - \int_0^t q_s ds,$$

is a G -Brownian motion under $\tilde{\mathbb{E}}$. Hence, X is a symmetric G -martingale under $\tilde{\mathbb{E}}$.

Using the first step, we now prove the assertion. If $\tilde{P}(T)$ is a symmetric G -martingale under $\tilde{\mathbb{E}}$,

$$\tilde{P}_t(T) = \tilde{\mathbb{E}}_t[\tilde{P}_T(T)] = \tilde{\mathbb{E}}_t[M_T^{-1}].$$

Since X is also a symmetric G -martingale under $\tilde{\mathbb{E}}$ and $A(T, T) = 0 = B(T, T)$, we have

$$X_t = \tilde{\mathbb{E}}_t[X_T] = \tilde{\mathbb{E}}_t[M_T^{-1}].$$

Thus, we have $\tilde{P}_t(T) = X_t$, which is equivalent to (18). Conversely, if (18) holds, we get $\tilde{P}_t(T) = X_t$ and hence, $\tilde{P}(T)$ is a symmetric G -martingale by the first step of the proof. ■

We use the process q to obtain an arbitrage-free term structure, since it serves as an adjustment factor for the uncertain volatility. The proof of Theorem 4.1 shows that the discounted bond cannot be a symmetric G -martingale under $\tilde{\mathbb{E}}$ in the presence of volatility uncertainty, since the expectation of the discount factor is not the same for every measure in the set of beliefs. The expectation differs among the measures in $\mathcal{P}$, since the short rate has a different variance under each of them, as Theorem 2.1 shows. So in order to unify the expectation of the discount factor under each measure, we need to adjust the short rate by the uncertainty about its variance. The following identity shows that the process q is a suitable adjustment factor, since it contains the same information as the variance of the short rate. By Proposition 2.1 and Proposition 3.4.5 of Peng (2019), we have

$$\begin{aligned} \hat{\mathbb{E}}[(r_t - \hat{\mathbb{E}}[r_t])^2] &= \hat{\mathbb{E}} \left[\left(\int_0^t e^{-\alpha(t-s)} dB_s \right)^2 \right] \\ &= \hat{\mathbb{E}} \left[\int_0^t e^{-2\alpha(t-s)} d\langle B \rangle_s \right] = \hat{\mathbb{E}}[q_t]. \end{aligned}$$

Therefore, we set $\lambda = q$ and use the Girsanov transformation for G -Brownian motion to adjust the short rate by its variance. Then the short rate evolves according to the dynamics

$$dr_t = (\mu(t) - \alpha r_t + q_t) dt + d\bar{B}_t.$$

Another important observation, which can be deduced from (17), is that the process q mean reverts twice as fast as the short rate towards the quadratic variation of the G -Brownian motion and thus, towards the quadratic variation

of the short rate. So the process always adjusts towards the correct belief about the volatility, which is unknown beforehand.

From an economic point of view, setting $\lambda = q$ is reasonable as well. In equation (20) we see that the instantaneous excess return of a zero-coupon bond with maturity T over the money-market account at time t is given by $B(t, T)q_t$. Dividing by the diffusion coefficient $-B(t, T)$, we obtain the market price of risk $-q_t$. In general, the market price of risk measures how much better we are doing with a bond compared to investing in the money-market account per one unit of risk. Since q is positive, we use a negative market price of risk. This is appropriate, because the bonds are not risky in this model. They have a certain payoff of 1 at the maturity, i.e. there is no default risk. On the other hand, investing in the money-market account is risky, since the short rate is stochastic and uncertain. Hence, we use a process representing the variance of the short rate to measure the risk and the uncertainty of the money-market account. So one may also refer to $-q$ as the *market price of uncertainty*.

In order to compare the bond prices from Theorem 6.1 with the prices from the traditional model, we derive an adjustment factor, linking both expressions. We denote the bond price from the traditional Hull–White model with constant volatility σ by $P_t^\sigma(T)$, which is defined by

$$P_t^\sigma(T) := \exp(A^\sigma(t, T) - B(t, T)r_t),$$

where $A^\sigma(t, T)$ and $B(t, T)$ are defined by (10a) and (10b), respectively. The bond price of the present model $P_t(T)$ is given by (18). Then we have

$$\frac{P_t(T)}{P_t^\sigma(T)} = \exp\left(-\int_t^T \frac{1}{2}\sigma^2 B(s, T)^2 ds - \frac{1}{2}B(t, T)^2 q_t\right).$$

The expression on the right-hand side represents an adjustment factor, which we can use to migrate from the traditional model to the model with volatility uncertainty.

Examining the adjustment factor, we note the following differences between the traditional and the present model. Since the adjustment factor is less than one, the bond prices in the present model are less than the prices in the classical model without volatility uncertainty. Moreover, we see that the squared term, depending on the volatility σ , is missing in $P_t(T)$. Instead, we have an additional term in $P_t(T)$, depending on the market price of uncertainty. Thus, the prices are independent of the volatility as well as the bounds for the volatility, which is the case in most of the models dealing with pricing under volatility uncertainty. It also implies that the bond price at the initial time corresponds to the price in the deterministic version of the Hull–White model without white noise, since the additional part in the exponential vanishes at the initial time. Though, this also applies to the standard Hull–White model after fitting it to the initial yield curve. In contrast to classical affine models, the bond price is now affine with respect to the short rate and the market price of uncertainty. The affine structure is similar to short rate models with a stochastic volatility (Fong and Vasicek 1991, Longstaff and Schwartz 1992). However, the additional factor in the bond price is not the volatility but a process adjusting towards the

current value of the quadratic variation of the short rate. A surprisingly similar structure can be found in the short rate model from Casassus *et al.* (2005), displaying unspanned stochastic volatility.

The most important implications of the prices in this model are as follows. Primarily, we manage to obtain a term structure which is robust with respect to the volatility itself and the bounds for the volatility. So we do neither have to estimate the volatility of the short rate in the future nor its bounds. Admittedly, there is a price we have to pay for this. We have to specify the market price of uncertainty, which appears in the bond prices. The market price of uncertainty depends on the past evolution of the quadratic variation of the G -Brownian motion, which corresponds to the quadratic variation of the short rate in the Hull–White model. The past evolution of the quadratic variation of the short rate is observable and can be inferred from market data. Alternatively, one could also estimate the variance of the short rate as an approximation for the market price of uncertainty. Moreover, in accordance to the short rate model of Casassus *et al.* (2005), the bond prices are completely unaffected by the structure of the volatility. The bounds of the volatility probably enter the model when it is used for pricing derivatives on bonds, that is, nonlinear contracts. Therefore, the model, despite its simple structure, could be a contribution to the literature on unspanned stochastic volatility, which was introduced by Collin-Dufresne and Goldstein (2002). A detailed discussion and the pricing of derivatives is, however, left for a subsequent study.

7. Yield curve fitting

As in the classical Hull–White model, we can use the time-dependent mean reversion level to fit the theoretical bond prices to an initially observable term structure. We introduce the following notions and assumptions, which are common in term structure models. Let us assume that there is an initial forward curve $f_0^* : [0, \tau] \rightarrow \mathbb{R}$, which is observed on the market. We assume that the initial forward curve f_0^* is differentiable and satisfies $f_0^*(0) = r_0$. The theoretical forward rate of the model is denoted by $f_t(T)$ for $t \leq T \leq \tau$ and defined by

$$f_t(T) := -\partial_T \log P_t(T).$$

The following theorem gives a necessary and sufficient condition for the theoretical forward curve matching the observable one at inception, which characterizes the mean reversion level of the short rate.

THEOREM 7.1 *Let the bond price be given by (18). Then it holds $f_0^*(T) = f_0(T)$ for all $T \leq \tau$ if and only if the mean reversion level of the short rate, for all $t \leq \tau$, satisfies*

$$\mu(t) = \alpha f_0^*(t) + \partial_t f_0^*(t). \quad (21)$$

Proof Let the bond price be given by (18). Then we can derive the initial forward rate $f_0(T)$. Taking the derivative of the logarithm of the bond price at time 0 and changing the

sign, we obtain

$$f_0(T) = \int_0^T \mu(t) e^{-\alpha(T-t)} dt + e^{-\alpha T} r_0. \quad (22)$$

Let us suppose that $f_0^*(T) = f_0(T)$. By (22), we then have

$$e^{\alpha T} f_0^*(T) = \int_0^T \mu(t) e^{\alpha t} dt + r_0.$$

Differentiating the latter equation with respect to T yields

$$\alpha e^{\alpha T} f_0^*(T) + e^{\alpha T} \partial_T f_0^*(T) = \mu(T) e^{\alpha T}.$$

Hence, the mean reversion level satisfies (21).

If we suppose that the mean reversion level satisfies (21), we can plug it into (22) and check, by reversing the above calculations, that it holds $f_0^*(T) = f_0(T)$. ■

After fitting the yield curve, the model is consistent with the classical Hull–White model. In general, the model is not consistent with the traditional Hull–White model, since the short rate dynamics and the bond prices differ from the ones in the traditional model, even if there is no volatility uncertainty. The adjusted short rate dynamics are given by

$$dr_t = (\mu(t) + q_t - \alpha r_t) dt + d\bar{B}_t,$$

which clearly differ from the Hull–White short rate dynamics. The bond prices also differ from the ones obtained in the Hull–White model, as the comparison from the previous section shows. However, the model becomes consistent with the classical one after fitting the model to the yield curve. After inserting (21) and the definition of q in the short rate dynamics, one can check that these are the same dynamics as in the fitted Hull–White model if there is no volatility uncertainty, i.e. if $\bar{\sigma} = \sigma = \underline{\sigma}$ (Brigo and Mercurio 2001, Subsection 3.3.1). Furthermore, inserting (21) in $A(t, T)$, which is defined in (19), and performing some calculations, yields

$$A(t, T) = - \int_t^T f_0^*(s) ds + f_0^*(t) B(t, T).$$

Plugging the above expression into (18) and dividing by the fitted bond price from the traditional Hull–White model with volatility σ (Brigo and Mercurio 2001, Subsection 3.3.2) leads to

$$\frac{P_t(T)}{P_t^\sigma(T)} = \exp \left(\frac{1}{2} B(t, T)^2 \left(\frac{\sigma^2}{2\alpha} (1 - e^{-2\alpha t}) - q_t \right) \right).$$

Thus, the adjustment factor is now determined by the difference between the variance of the short rate with constant volatility and the uncertain variance of the short rate with volatility uncertainty. Hence, the adjustment factor is equal to 1 if $\bar{\sigma} = \sigma = \underline{\sigma}$.

The consistency with the classical Hull–White model further justifies the choice of q as the market price of uncertainty. The discussion from above shows that the adjustment factor q appearing in the short rate dynamics is actually included in

the dynamics of the classical model after fitting the theoretical prices to the observed ones. Hence, the process q naturally appears in the risk-neutral dynamics of the short rate, which gives another justification for choosing the market price of uncertainty in this particular way. The interesting thing, however, is that in the classical model, this expression is used for yield curve fitting, whereas in this model, the expression is needed in order to have an arbitrage-free model.

In order to completely calibrate the model, one has to establish a robust estimation procedure for the mean reversion speed. Theorem 7.1 characterizes the mean reversion level in terms of an initially observable term structure. However, the term structure still involves a parameter: the mean reversion speed α . A typical approach in short rate models to estimate parameters is to use a maximum likelihood approach. The maximum likelihood approach heavily relies on the probabilistic law of the short rate. In the presence of volatility uncertainty, there is a family of possible probabilistic laws for the short rate and we are uncertain about which one is correct. Therefore, one has to use a robust approach to calibrate the model instead of the classical maximum likelihood approach.

8. Multifactor extension

The previous model can be generalized to a model driven by multiple factors. For this purpose, we consider a d -dimensional G -Brownian motion $B = (B^1, \dots, B^d)$. That is, B^i is a 1-dimensional G -Brownian motion for $i = 1, \dots, d$. A d -dimensional G -Brownian motion can be constructed as in Section 2 by replacing $C_0(\mathbb{R}_+)$ and $[\underline{\sigma}, \bar{\sigma}]$ with $C_0^d(\mathbb{R}_+)$, the space of all $\mathbb{R}^d$ -valued continuous paths on $\mathbb{R}_+$ starting in 0, and a bounded, closed, and convex subset $\Sigma \subset \mathbb{R}^{d \times d}$, respectively. Hence, we allow for a possible uncertain correlation between the risk factors. The short rate process is defined by

$$r_t := \mu(t) + \sum_{i=1}^d X_t^i, \quad (23)$$

where $\mu : \mathbb{R}_+ \rightarrow \mathbb{R}$ is a suitably integrable function and, for $i = 1, \dots, d$, the factor X^i satisfies

$$dX_t^i = -\alpha_i X_t^i dt + dB_t^i, \quad X_0^i = 0$$

for some constant $\alpha_i > 0$. The process X^i , for $i = 1, \dots, d$, is given by

$$X_t^i = \int_0^t e^{-\alpha_i(t-s)} dB_s^i$$

and represents a risk factor that affects the short rate.

Such a multifactor extension does not lead to an arbitrage-free term structure. Similar to the discussion in Section 4, we can show that the short rate dynamics given by (23) are not suitable for martingale modeling. As in Theorem 4.1, we can show that the discounted bonds can be symmetric G -martingales under $\hat{\mathbb{E}}$ if and only if there is no volatility uncertainty, that is, Σ is a singleton. This can be shown, as in the proof of Theorem 4.1, by considering two different beliefs about the volatility, which lead to different bond prices.

We consider sublinear expectations defined by a G -backward stochastic differential equation and equivalent to the initial sublinear expectation in order to find an arbitrage-free term structure. Let $\lambda = (\lambda^1, \dots, \lambda^d)$ be a d -dimensional bounded process in $M_G^p(0, \tau; \mathbb{R}^d)$ for some $p > 1$. We define the sublinear expectation $\bar{\mathbb{E}}$ on $L_G^p(\Omega_\tau)$ by $\bar{\mathbb{E}}_t[\xi] := Y_t^\xi$, where Y^ξ solves the G -backward stochastic differential equation

$$-Y_t^\xi = \sum_{i=1}^d \lambda_t^i Z_t^i dt - \sum_{i=1}^d Z_t^i dB_t^i - dK_t, \quad Y_\tau^\xi = \xi.$$

Then we can show, as in Proposition 5.1, that the bond market is arbitrage-free if the discounted bonds are symmetric G -martingales under $\bar{\mathbb{E}}$.

We obtain an arbitrage-free term structure by considering a particular sublinear expectation of the above form. Let us define the process $q = (q^1, \dots, q^d)$ by

$$q_t^i := \sum_{j=1}^d q_t^{ij}$$

for $i = 1, \dots, d$, where q_t^{ij} , for $j = 1, \dots, d$, is defined by

$$q_t^{ij} := \int_0^t e^{-(\alpha_i + \alpha_j)(t-s)} d\langle B^i, B^j \rangle_s.$$

By applying Itô's formula for G -Brownian motion, we know that q_t^{ij} satisfies

$$dq_t^{ij} = d\langle B^i, B^j \rangle_t - (\alpha_i + \alpha_j) q_t^{ij} dt, \quad q_0^{ij} = 0.$$

If we use the process q to define a sublinear expectation as above, then there is a unique arbitrage-free term structure.

THEOREM 8.1 *Let $T \leq \tau$ and $\lambda = q$. Then the discounted bond $\tilde{P}(T)$ is a symmetric G -martingale under $\bar{\mathbb{E}}$ if and only if the bond price, for all $t \leq T$, is given by*

$$P_t(T) := \exp \left(- \int_t^T \mu(s) ds - \sum_{i=1}^d B_i(t, T) X_t^i - \frac{1}{2} \sum_{i,j=1}^d B_i(t, T) B_j(t, T) q_t^{ij} \right),$$

where the function $B_i : \mathbb{R}^2 \rightarrow \mathbb{R}$, for $i = 1, \dots, d$, is defined by

$$B_i(t, T) := \frac{1}{\alpha_i} (1 - e^{-\alpha_i(T-t)}).$$

Proof Let $T \leq \tau$ and $\lambda = q$. As in the proof of Theorem 6.1, the assertion follows if we show that the process X , defined by

$$X_t := \exp \left(- \int_0^T \mu(s) ds - \sum_{i=1}^d B_i(t, T) X_t^i - \frac{1}{2} \sum_{i,j=1}^d B_i(t, T) B_j(t, T) q_t^{ij} - \sum_{i=1}^d \int_0^t X_s^i ds \right),$$

is a symmetric G -martingale. Applying Itô's formula for G -Brownian motion to X yields

$$dX_t = \Delta_t X_t dt - \sum_{i=1}^d B_i(t, T) X_t dB_t^i + \sum_{i,j=1}^d \Delta_t^{ij} X_t d\langle B^i, B^j \rangle_t,$$

where the drift terms Δ and Δ^{ij} , for $i, j = 1, \dots, d$, are given by

$$\begin{aligned} \Delta_t &:= - \sum_{i=1}^d \partial_t B_i(t, T) X_t^i - \frac{1}{2} \sum_{i,j=1}^d (\partial_t B_i(t, T) B_j(t, T) \\ &\quad + B_i(t, T) \partial_t B_j(t, T)) q_t^{ij} \\ &\quad - \sum_{i=1}^d B_i(t, T) (-\alpha_i X_t^i) - \frac{1}{2} \sum_{i,j=1}^d B_i(t, T) B_j(t, T) \\ &\quad \times (-\alpha_i + \alpha_j) q_t^{ij} - \sum_{i=1}^d X_t^i \\ &= - \sum_{i=1}^d (\partial_t B_i(t, T) - \alpha_i B_i(t, T) + 1) X_t^i \\ &\quad - \frac{1}{2} \sum_{i,j=1}^d B_j(t, T) (\partial_t B_i(t, T) - \alpha_i B_i(t, T)) q_t^{ij} \\ &\quad - \frac{1}{2} \sum_{i,j=1}^d B_i(t, T) (\partial_t B_j(t, T) - \alpha_j B_j(t, T)) q_t^{ij}, \\ \Delta_t^{ij} &:= -\frac{1}{2} B_i(t, T) B_j(t, T) + \frac{1}{2} B_i(t, T) B_j(t, T) = 0, \end{aligned}$$

respectively. Since the function B_i , for $i = 1, \dots, d$, satisfies

$$\partial_t B_i(t, T) = \alpha_i B_i(t, T) - 1$$

and $q^{ij} = q^{ji}$ for $i, j = 1, \dots, d$, we obtain

$$dX_t = \sum_{i=1}^d B_i(t, T) X_t q_t^i dt - \sum_{i=1}^d B_i(t, T) X_t dB_t^i.$$

We can use the same argument as in the proof of Theorem 6.1 to show that $X_t \in L_G^2(\Omega_t)$ for all $t \leq T$. By the Girsanov transformation for G -Brownian motion, the process $\bar{B} = (\bar{B}^1, \dots, \bar{B}^d)$, defined by

$$\bar{B}_t^i := B_t^i - \int_0^t q_s^i ds$$

for $i = 1, \dots, d$, is a G -Brownian motion under $\bar{\mathbb{E}}$. Therefore, the process X is a symmetric G -martingale under $\bar{\mathbb{E}}$. ■

We can use the function μ to fit the model to an initially observed term structure. Let us assume that there is a sufficiently regular forward curve f_0^* , which is currently observed on the market. Then it is easy to check that the theoretical forward curve implied by the model matches the observed one at inception if and only if $\mu(t) = f_0^*(t)$ for all $t \leq \tau$.

9. Conclusion

In the present paper, we investigate the traditional Hull–White model when there is Knightian uncertainty about the volatility. We show that the common approach to pricing zero-coupon bonds, martingale modeling, does not work in the presence of volatility uncertainty. Hence, we follow a different approach. By introducing a market price of uncertainty, we adjust the short rate by its uncertain variance to obtain an arbitrage-free term structure. The resulting term structure is completely robust with respect to the volatility. The bond prices neither depend on the future evolution of the volatility nor on its bounds. Instead, they depend on the current value of the market price of uncertainty. In particular, the bonds are exponentially affine with respect to the short rate and the market price of uncertainty. Due to the adjustment of the short rate, the model is inconsistent with the traditional Hull–White model. However, the model becomes consistent with the traditional one after fitting the model prices to the yield curve. All of these results hold true if the short rate is driven by multiple risk factors.

A first question for further research is whether the proposed term structure and the particular choice of the market price of uncertainty can be supported by an equilibrium in a representative agent economy. The present approach is purely based on no-arbitrage pricing. Instead, one could investigate a structural model in the spirit of Cox *et al.* (1985) under model uncertainty. Gagliardini *et al.* (2009) examine a structural model with ambiguity. The representative agent in the model faces ambiguity about the drift of the underlying risk factors. Since she is ambiguity averse, the agent has to solve a max-min expected utility problem. The solution determines the uncertain drift process, which is termed the *market price of ambiguity*. In a similar fashion, one could examine a structural model in which the representative agent faces ambiguity about the volatility. For this purpose, one could adapt the framework of Epstein and Ji (2013) to find out if there is an equilibrium in a representative agent economy supporting the specific market price of uncertainty used in the specified model.

Apart from that, it would be interesting to test the empirical performance of the model, especially in comparison to traditional term structure models. One could, e.g. test if the model is able to explain the violation of the expectations hypothesis as is done, for instance, by Dai and Singleton (2003) and Gagliardini *et al.* (2009) and whether it does better than traditional models. Fama and Bliss (1987), Campbell and Shiller (1991), and Cochrane and Piazzesi (2005) empirically test the expectations hypothesis by regressing changes in the yield curve onto the slope of the yield curve, which shows that the expectations hypothesis is violated. The regression produces negative coefficients, decreasing with respect to the maturity. Dai and Singleton (2003) test the ability of several term structure models to explain these empirical findings. For this purpose, they fit the models to data and simulate term structures from the fitted models. Then they run regressions as above and compare the regression coefficients from the simulated data with the ones from real data. A successful model should match the coefficients from real data. In order to test the performance of the present model in this regard, one has

to use a robust approach to calibrate the model as is described at the end of Section 7. Moreover, one has to develop a simulation procedure which works in the presence of volatility uncertainty. Volatility is uncertain in the sense of Knightian uncertainty. By definition, Knightian uncertainty cannot be measured by any probability. Thus, standard simulation procedures cannot be used. Instead, one has to construct a robust simulation procedure.

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